Lecture 1: Linear Algebra for Machine Learning

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Outline

1. Scalar, vector, matrix and tensor

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2. System of equations

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3. Norms

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4. Special Matrices and Vectors

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5. Eigenvectors and eigenvalues

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6. Singular Value Decomposition

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3. Norms
4. Special Matrices and Vectors
5. Eigenvectors and eigenvalues
6. Singular Value Decomposition
7. Moore-Penrose Pseudoinverse

Scalars

- A scalar is a single number
- Integers, real numbers, rational numbers, etc.
- We denote it with italic font:

$$a, \quad n, \quad x$$

Vectors

- A vector is a 1-D array of numbers:

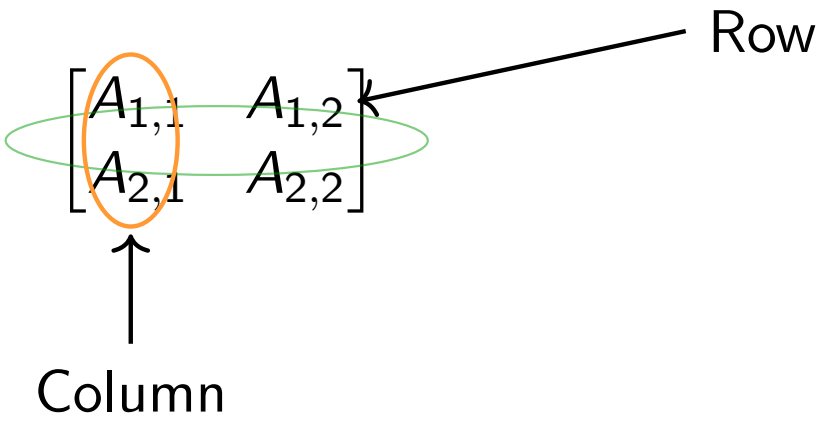
$$\boldsymbol{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

- Can be real, binary, integer, etc.
- Example notation for type and size:

$$\boldsymbol{x} \in \mathbb{R}^n$$

Matrices

- A matrix is a 2-D array of numbers:



- Example notation for type and shape:

$$\boldsymbol{A} \in \mathbb{R}^{m \times n}$$

Matrices

- A matrix is a 2-D array of numbers:
- Example notation for type and shape:

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^\top = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

$$(\boldsymbol{A}^\top)_{i,j} = A_{j,i}$$

Tensors

- A tensor is an array of numbers, that may have
  - zero dimensions, and be a scalar
  - one dimension, and be a vector
  - two dimensions, and be a matrix
  - or more dimensions.

Matrix Transpose

$$\boldsymbol{A} = \begin{bmatrix} A_{1,1} & A_{1,2} \\ A_{2,1} & A_{2,2} \\ A_{3,1} & A_{3,2} \end{bmatrix} \Rightarrow \boldsymbol{A}^\top = \begin{bmatrix} A_{1,1} & A_{2,1} & A_{3,1} \\ A_{1,2} & A_{2,2} & A_{3,2} \end{bmatrix}$$

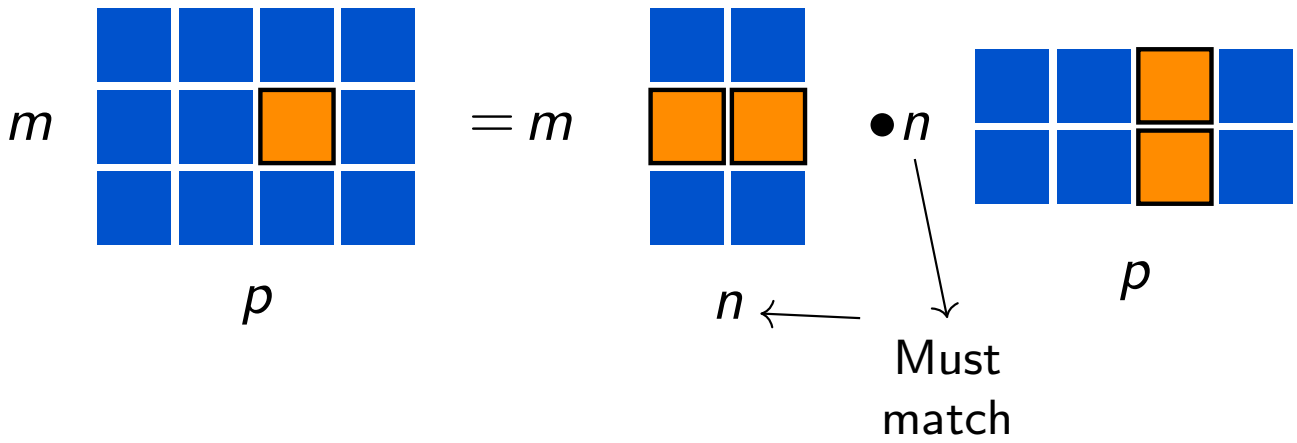
Figure 2.1: The transpose of the matrix can be thought of as a mirror image across the main diagonal.

$$(\boldsymbol{A}^\top)_{i,j} = A_{j,i} \qquad (\boldsymbol{AB})^\top = \boldsymbol{B}^\top \boldsymbol{A}^\top$$

Matrix (Dot) Product

$$\boldsymbol{C} = \boldsymbol{AB}. \tag{1}$$

$$C_{i,j} = \sum_k A_{i,k} B_{k,j}. \tag{2}$$



Identity Matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Figure 2.2: Example identity matrix: This is $\boldsymbol{I}_3$.

$$\forall \boldsymbol{x} \in \mathbb{R}^n, \quad \boldsymbol{I}_n \boldsymbol{x} = \boldsymbol{x}.$$

Systems of Equations

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b} \tag{3}$$

expands to

$$\boldsymbol{A}_{1,:}\boldsymbol{x} = b_1 \tag{4}$$
$$\boldsymbol{A}_{2,:}\boldsymbol{x} = b_2 \tag{5}$$
$$\vdots$$
$$\boldsymbol{A}_{m,:}\boldsymbol{x} = b_m \tag{6}$$

Systems of Equations

$$A_{1,1}x_1 + A_{1,2}x_2 + \cdots + A_{1,n}x_n = b_1 \tag{7}$$
$$A_{2,1}x_1 + A_{2,2}x_2 + \cdots + A_{2,n}x_n = b_2 \tag{8}$$
$$\vdots$$
$$A_{m,1}x_1 + A_{m,2}x_2 + \cdots + A_{m,n}x_n = b_m \tag{9}$$

Solving Systems of Equations

- A linear system of equations can have:
  - No solution
  - Many solutions
  - Exactly one solution: this means multiplication by the matrix is an invertible function

Matrix Inversion

- Matrix inverse:

$$\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}_n$$

- Solving a system using an inverse:

$$\begin{aligned} \boldsymbol{A}\boldsymbol{x} &= \boldsymbol{b} \\ \boldsymbol{A}^{-1}\boldsymbol{A}\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \\ \boldsymbol{I}_n\boldsymbol{x} &= \boldsymbol{A}^{-1}\boldsymbol{b} \end{aligned}$$

- Numerically unstable, but useful for abstract analysis

Invertibility

- Matrix can't be inverted if...
  - More rows than columns
  - More columns than rows
  - Redundant rows/columns ("linearly dependent", "low rank")

Norms

- Functions that measure how "large" a vector is
- Similar to a distance between zero and the point represented by the vector

$$\|\boldsymbol{x}\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- A norm is any function f satisfying:
  - $f(\boldsymbol{x}) = 0 \Rightarrow \boldsymbol{x} = \boldsymbol{0}$
  - $f(\boldsymbol{x} + \boldsymbol{y}) \leq f(\boldsymbol{x}) + f(\boldsymbol{y})$ (the triangle inequality)
  - $\forall \alpha \in \mathbb{R},\ f(\alpha \boldsymbol{x}) = |\alpha| f(\boldsymbol{x})$

Norms

- L^p norm

$$\|\boldsymbol{x}\|_p = \left(\sum_i |x_i|^p \right)^{\frac{1}{p}}$$

- Most popular norm: L^2 norm, $p = 2$
- L^1 norm, $p = 1$:

$$\|\boldsymbol{x}\|_1 = \sum_i |x_i|.$$

- Max norm, infinite p:

$$\|\boldsymbol{x}\|_\infty = \max_i |x_i|.$$

- Frobenius norm (for matrices):

$$\|\boldsymbol{A}\|_F = \sqrt{\sum_{i,j} A_{i,j}^2}$$

Special Matrices and Vectors

- Unit vector:

$$\|\boldsymbol{x}\|_2 = 1.$$

- Symmetric Matrix:

$$\boldsymbol{A} = \boldsymbol{A}^\top.$$

- Orthogonal matrix:

$$\boldsymbol{A}^\top \boldsymbol{A} = \boldsymbol{A}\boldsymbol{A}^\top = \boldsymbol{I}$$
$$\boldsymbol{A}^{-1} = \boldsymbol{A}^\top$$

Eigendecomposition

- Eigenvector and eigenvalue:

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v}.$$

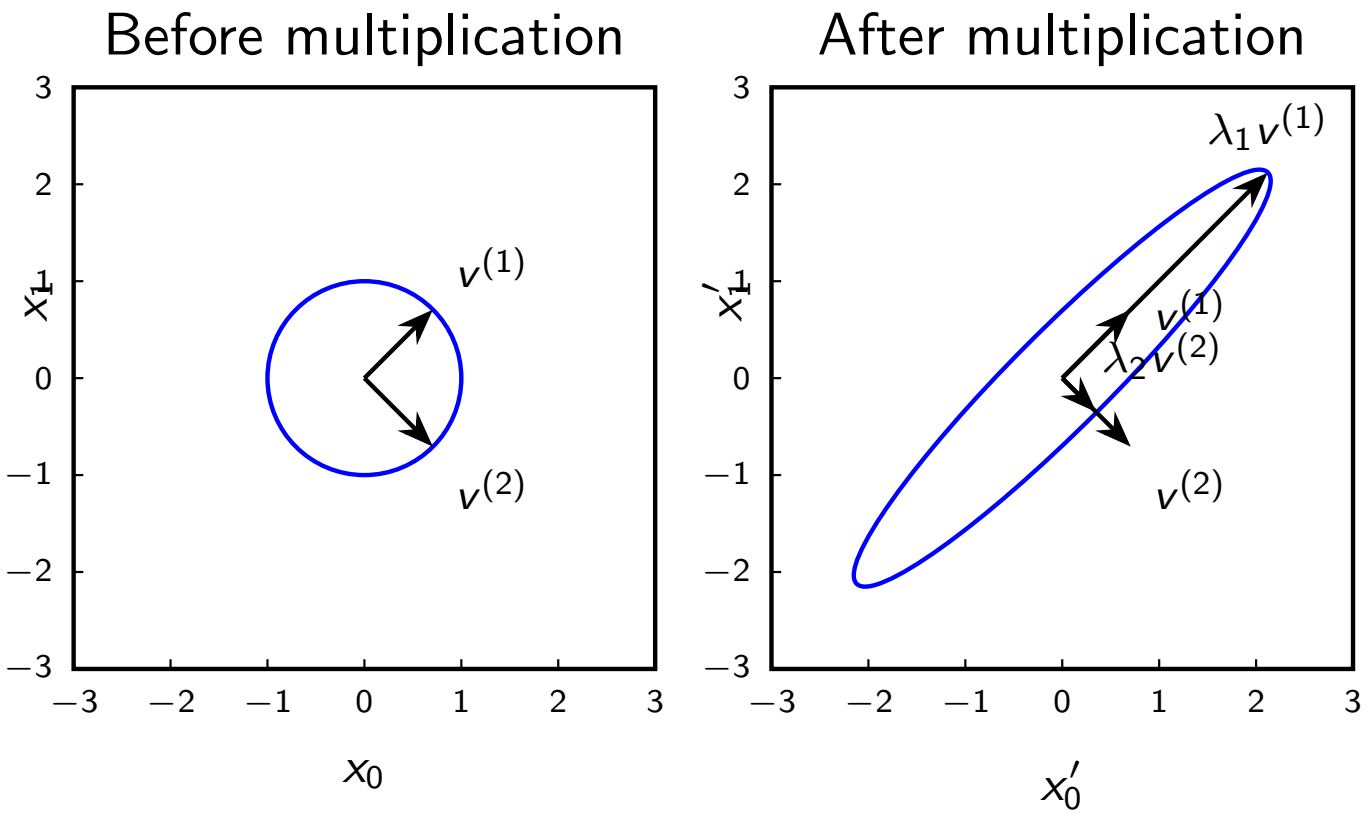
- Eigendecomposition of a diagonalizable matrix:

$$\boldsymbol{A} = \boldsymbol{V}\,\mathrm{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}.$$

- Every real symmetric matrix has a real, orthogonal eigendecomposition:

$$\boldsymbol{A} = \boldsymbol{Q}\boldsymbol{\Lambda}\boldsymbol{Q}^{\top}$$

Effect of Eigenvalues



Singular Value Decomposition

- Similar to eigendecomposition
- More general; matrix need not be square

$$\text{Eigendecomposition:} \quad \boldsymbol{A} = \boldsymbol{V}\,\text{diag}(\boldsymbol{\lambda})\boldsymbol{V}^{-1}$$

$$\text{SVD:} \quad \boldsymbol{A} = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\top}$$

If $\boldsymbol{A}$ is $m \times n$, then $\boldsymbol{U}$ is $m \times m$, $\boldsymbol{D}$ is $m \times n$, and $\boldsymbol{V}$ is $n \times n$.

Moore-Penrose Pseudoinverse

A^+ is the unique matrix satisfying

$$AA^+A = A, \tag{10}$$
$$A^+AA^+ = A^+, \tag{11}$$
$$(AA^+)^T = AA^+, \tag{12}$$
$$(A^+A)^T = A^+A. \tag{13}$$

- If the equation

$$\boldsymbol{x} = \boldsymbol{A}^+\boldsymbol{y}$$

has:

  - Exactly one solution: this is the same as the inverse.
  - No solution: this gives us the solution with the smallest error $\|\boldsymbol{A}\boldsymbol{x} - \boldsymbol{y}\|_2$.
  - Many solutions: this gives us the solution with the smallest norm of $\boldsymbol{x}$.

Useful properties

- $(A^+)^+ = A$.
- $(A^T)^+ = (A^+)^T$.
- If A is square and invertible, then $A^+ = A^{-1}$.
- For a scalar or a diagonal matrix, invert the nonzero entries and leave the zeros alone.
- **Caution:** $(AB)^+ \neq B^+A^+$ in general.

Let

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

(rank 1). Then

$$A^{+} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

The ordinary inverse does not exist, but the pseudoinverse inverts the part that can be inverted and ignores the rest.
If we want to solve $Ax = b$, with $b = (1, 1)^T$, then

$$x^* = A^{+}b = \arg\min \|Ax - b\|_2$$

Example

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}.$$

It has full row rank , so we can use $A^+ = A^T(AA^T)^{-1}$.

$$AA^T = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (AA^T)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

$$A^+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \cdot \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \\ 1 & 1 \end{pmatrix}.$$

- $AA^+ = I_2$, since A has full row rank, so A^+ is a right inverse.

Example

Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Since $AB = I_2$, we have

$$(AB)^+ = I_2.$$

The Moore–Penrose pseudoinverses are

$$A^+ = A^{\mathsf{T}}(AA^{\mathsf{T}})^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}, \qquad B^+ = (B^{\mathsf{T}}B)^{-1}B^{\mathsf{T}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Therefore,

$$\boxed{(AB)^+ = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \neq \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} = B^+A^+.}$$

Computing the Pseudoinverse

The SVD allows the computation of the pseudoinverse:

$$\boldsymbol{A}^{+} = \boldsymbol{V}\boldsymbol{D}^{+}\boldsymbol{U}^{\top}$$

Take reciprocal of non-zero entries

Equivalently, the pseudoinverse is defined as

$$\boldsymbol{A}^{+} = \lim_{\alpha \searrow 0} (\boldsymbol{A}^{\top}\boldsymbol{A} + \alpha \boldsymbol{I})^{-1}\boldsymbol{A}^{\top}$$

where $\boldsymbol{U}, \boldsymbol{D}, \boldsymbol{V}$ are the SVD of $\boldsymbol{A}$ and $\boldsymbol{D}^{+}$ is obtained by taking the reciprocal of the non-zero elements of $\boldsymbol{D}$ and then transposing the result.

Trace

$$\text{Tr}(\boldsymbol{A}) = \sum_i A_{i,i}.$$

$$\text{Tr}(\boldsymbol{A}) = \text{Tr}(\boldsymbol{A}^\top)$$

$$\text{Tr}(\boldsymbol{ABC}) = \text{Tr}(\boldsymbol{CAB}) = \text{Tr}(\boldsymbol{BCA})$$

Learning linear algebra

- Do a lot of practice problems
- Start out with lots of summation signs and indexing into individual entries
- Eventually you will be able to mostly use matrix and vector product notation quickly and easily